Module 14: Macroevolution - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. State the biological species concept and its limits.
2. Explain reproductive isolation as reduced gene flow.
3. Classify examples as prezygotic or postzygotic barriers.
4. Compare allopatric and sympatric speciation.
5. Interpret speciation as population-level divergence over time.

Topics

- Species concepts
- Reproductive isolation
- Prezygotic barriers
- Postzygotic barriers
- Speciation patterns and phylogeny

Key Terms to Know

- **Species** - Lineage or group identified by a species concept.
- **Biological species concept** - Species defined by interbreeding and reproductive isolation.
- **Reproductive isolation** - Barriers that reduce gene flow between groups.
- **Prezygotic barrier** - Barrier before fertilization.
- **Postzygotic barrier** - Barrier after fertilization.
- **Allopatric speciation** - Speciation with geographic separation.
- **Sympatric speciation** - Speciation without geographic separation.
- **Phylogeny** - Evolutionary relationship history among lineages.

Core Contents

1. The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.
2. Reproductive isolation reduces gene flow. Appearance or separation alone does not establish species boundaries; some distinct species hybridize.
3. Prezygotic barriers act before fertilization: habitat, timing, behavior, anatomy, or gamete incompatibility can prevent a zygote.
4. Postzygotic barriers reduce hybrid development, survival, or reproduction. Classify the stage using the evidence provided.
5. Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.

Study Tips

1. Complete Lab 15 as a prediction, evidence check, and revision exercise. No internet or special equipment is needed.
2. Answer the learning questions as homework practice; show reasoning and units when a question supplies numbers.
3. Use Practice Test 03, then check its self-contained key and explain any correction in your own words.
4. Connect each answer to a mechanism and evidence, rather than a memorized definition alone.

Connected Lab

- `lab-15_macroevolution.md`

Generated Visuals

- Module 14: Macroevolution Evidence Map
- Module 14: Macroevolution Reasoning Sequence
- Module 14: Macroevolution Retrieval and Lab Check